Answer key - Stationary Points and Optimisation

GCE Advanced Level - Mathématiques - Corrige

Detailed method for every exercise. Follow the substitutions line by line.
Corrige detaille de chaque exercice. Suis les substitutions ligne par ligne.

1. For f(x) = x^2 - 6x + 5, find the x-coordinate where f'(x) = 0.
Pour f(x) = x^2 - 6x + 5, trouve l abscisse ou f'(x) = 0.

Answer: 3
f'(x) = 2x - 6 = 0 gives x = 3.
f'(x) = 2x - 6 = 0 donne x = 3.

2. A stationary point has f''(x) = -6 at that point. Is it a minimum or a maximum?
Un point stationnaire verifie f''(x) = -6 en ce point. Est-ce un minimum ou un maximum ?

- A maximum / Un maximum *
- A minimum / Un minimum
- A point of inflection / Un point d inflexion
- Cannot be decided / Impossible a decider

Answer: A maximum
A negative second derivative means the curve is concave down, so the point is a maximum.
Une derivee seconde negative signifie que la courbe est concave : le point est un maximum.

3. For f(x) = x^2 - 6x + 5, find the x-coordinate of the stationary point.
Pour f(x) = x^2 - 6x + 5, trouve l abscisse du point stationnaire.

Answer: 3
f'(x) = 2x - 6 = 0 gives x = 3.
f'(x) = 2x - 6 = 0 donne x = 3.

4. At a stationary point f''(x) = 4. Is it a minimum or a maximum?
En un point stationnaire f''(x) = 4. Est-ce un minimum ou un maximum ?

- A minimum / Un minimum *
- A maximum / Un maximum
- A point of inflection / Un point d inflexion
- Cannot be decided / Impossible a decider

Answer: A minimum
A positive second derivative means concave up, so the point is a minimum.
Une derivee seconde positive signifie convexe : le point est un minimum.

5. For f(x) = x^2 - 6x + 5, find the y-coordinate of the stationary point.
Pour f(x) = x^2 - 6x + 5, trouve l ordonnee du point stationnaire.

Answer: -4
At x = 3, f(3) = 9 - 18 + 5 = -4.
En x = 3, f(3) = 9 - 18 + 5 = -4.

6. What defines a stationary point of f?
Qu est-ce qui definit un point stationnaire de f ?

- f'(x) = 0 *
- f(x) = 0

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- f''(x) = 0
- x = 0

Answer: f'(x) = 0

A stationary point is where the gradient is zero, so f'(x) = 0.

Un point stationnaire est un point ou la pente est nulle, donc f'(x) = 0.

7. A rectangular pen uses 40 metres of fence with width x and length 20 - x. Find the width x that maximises the area.

Un enclos rectangulaire utilise 40 metres de cloture avec une largeur x et une longueur 20 - x. Trouve la largeur x qui maximise l aire.

Answer: 10

A(x) = 20x - x^2, so A'(x) = 20 - 2x = 0 gives x = 10. Since A''(x) = -2 < 0, this is a maximum.

A(x) = 20x - x^2, donc A'(x) = 20 - 2x = 0 donne x = 10. Comme A''(x) = -2 < 0, c est un maximum.

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